Spotify Playlist Curator Using Machine Learning

UE24CS352A — Machine Learning Mini-Project

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Abstract

This project builds a supervised multiclass classifier that predicts one of 13 selected playlist classes from track-level audio attributes and metadata. It is inspired by Awadelkarim and Coelho's paper, Training a Playlist Curator Based on User Taste. The implementation first evaluates paper-inspired Perceptron, support vector machine (SVM), and neural-network baselines using the ten audio features available in the supplied CSV. It then tests a richer feature representation and additional classifiers, including tree ensembles. A controlled feature-ablation experiment compares audio-only inputs with an extended set of track-level features. The work is an adaptation rather than an exact reproduction because the supplied dataset does not contain the paper's original curated playlists and complete

2,095-dimensional artist-genrerepresentation.

1. Problem Statement and Objectives

Playlist curation involves grouping songsaccording toa theme, mood, genre, or listening context. These relationships can be nonlinear and may not be captured by one musical measurement alone. The project formulates playlist assignment as supervised multiclass classification: given a feature vector for a song, predict its playlist label among 13 selected classes.

The objectives are to (1) clean and inspect the supplied Spotify tracks dataset, (2) implement models aligned with the reference paper, (3) evaluate whether richer track metadata and engineered features improve performance, and (4) reportresults using cross-validation, held-out testing, and class-sensitive metrics.

2. Dataset and Preparation

Thesupplied spotify_songs.csvcontains32,833 rows and 23 columns, including track identifiers and names, artist and album metadata, playlist identifiers/names, and audio attributes such as danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, and duration. The notebook selects 13 eligible playlists using a fixed random seed and applies documented cleaning to create a single-label classification subset. The resulting cleaned set described in the notebook contains 1,241 tracks across 13 classes. Playlist identifiers and other target-derived fields are excluded from model inputs to reduce target leakage.

Two input representations are evaluated. The paper-inspired representation uses ten audio features. The extended representation adds available track metadata and engineered variables, including transformed duration, release year/decade, popularity, categorical key encoding, and selected interactions among audio attributes. Numerical missing values are imputed; numerical and categorical preprocessing is fitted within pipelines to reduce leakage from validation/test data. Artist-grouped splitting is used so the same normalized artist is not present in both the training and held-out test sets.

3. Methodology and Implementation

Paper-inspired baselines include a one-versus-rest Perceptron and SVM classifiers with RBF, sigmoid, and polynomial kernels, each evaluated with and without feature scaling. Two PyTorch feed-forward neural networks are also trained: a one-hidden-layer network with sigmoid activation and a LogSoftmax output, and a two-hidden-layer comparison network. These experiments approximate the paper's model families on the available CSV; they do not reproduce its original genre-tag inputs.

The enhanced-feature benchmark evaluates Logistic Regression, Random Forest, Extra Trees, an RBF SVM, an MLP classifier, and HistGradientBoosting. Randomized hyperparameter searches are performed for HistGradientBoosting and Random Forest using grouped cross-validation on the training data. Equal-weight and weighted soft-voting ensembles combine the predictions of HistGradientBoosting, Random Forest, and Extra Trees. The notebook records model comparisons, a classification report, a confusion matrix, neural-network training curves, and feature-ablation results.

Evaluation uses five-fold artist-grouped cross-validation within the training data and a separate held-out artist-grouped test set. Reported metrics include accuracy and macro-F1. Accuracy is the fraction of correct predictions; macro-F1 averages class-wise F1 scores equally and therefore provides a more balanced view when classes have different sizes.The held-out test set should not be used to choose hyperparameters or repeatedly select a model.

4. Results

Model	Feature set	Test accuracy	Macro-F1
Equal soft-voting ensemble	Extended 31	77.82%	0.761
Weighted soft-voting ensemble	Extended 31	77.42%	0.757
Tuned HistGradientBoosting	Extended 31	76.61%	0.748
Tuned Random Forest	Extended 31	76.21%	0.755

In the recorded combined run, the equal soft-voting ensemble achieved 77.82% held-out test accuracy and a macro-F1 of 0.761. The weighted ensemble achieved 77.42%; tuned HistGradientBoosting achieved 76.61%; and tuned Random Forest achieved 76.21%. On the paper-inspired audio-only test comparison, the scaled RBF SVM achieved 42.34% accuracy. These results are from the adapted dataset and split, and should not be directly compared to the paper's reported 82% as if they were the same experiment.

The feature-ablation results show a marked improvement when the extended features are used. For example, HistGradientBoosting's mean grouped cross-validation accuracy rose from approximately 46.0% using the paper's ten audio features to approximately 73.1% using the extended feature set. This indicates that the richer representation is useful in this experiment; because several features are added together, it does not establish which individual feature caused the gain.

5. Conclusion, Limitations, and Future Work

Theproject demonstratesa completesupervised-learning workflow for playlist classification, from data cleaning and feature engineering to model tuning and evaluation. The enhanced features and tree-based ensembles perform substantially better than the paper-inspired audio-only baselines on this selected dataset. The work remains an adaptation: the paper used 1,044 examples from its own 13 curated playlists and 2,095 artist-genre one-hot features in addition to ten audio features, which are not available in the supplied CSV in the same form. Other limitations include simplifying songs with multiple playlist memberships to a single label and relying on historical dataset labels rather than human listening preference.

Future work could investigate multi-label classification, richer genre/artist representations, temporal validation on newer releases, nested cross-validation for model selection, and user-centred evaluation of playlist relevance, diversity,and satisfaction.

Reference

Awadelkarim, A., & Coelho, K. (2018). Training a Playlist Curator Based on User Taste. Supplied research paper (22-1.pdf).